

TOWARDS INCLUSIVE HEALTHCARE: LEVERAGING LOW-RESOURCE TRANSLATION FOR MULTILINGUAL COMMUNICATION

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01 Introduction

Machine Learning, NMT Systems, and LLMs

Neural Machine Translation (NMT) has revolutionized language translation through deep learning, particularly using sequence-to-sequence (Seq2Seq) architectures based on transformers.

- Requires substantial training data, creating challenges for low-resource languages like Vietnamese
- Pretrained multilingual models like MBart50 transfers knowledge from high-resource to low-resource settings
- Hugging Face Transformers library and Seq2SeqTrainer can be used to fine-tune MBart50 on domain-specific healthcare data

Low-Resource Languages in Healthcare Domains

In healthcare settings, language barriers can contribute to miscommunication, reduced care quality, and health disparities—especially for speakers of low-resource languages like Vietnamese.

- Neural Machine Translation (NMT) systems show strong performance for high-resource language pairs but often underperform when applied to languages with limited parallel data and specialized vocabulary
- Advancements in Large Language Models (LLMs), such as MBart50, present an opportunity to address these limitations through transfer learning and multilingual fine-tuning

Can a multilingual NMT system,
enhanced by LLM integration and fine-
tuned on low-resource, domain-
specific data, achieve effective and
accurate translation performance?

Research Question

Hypothesis

We hypothesize that leveraging pretrained LLMs and adapting them through sequence-to-sequence training on targeted datasets can significantly improve translation quality for low-resource languages.

Why does this matter?

In healthcare contexts where precision is critical, this research aims to provide scalable solutions for real-time, high-quality translation of medical content—enabling effective communication across language barriers and ultimately supporting more inclusive and accessible care for underrepresented linguistic communities.

02 Methodology

Two-Phase Experimental Approach

1. Establish French-English translation benchmarks using traditional and LLM-based models
2. Fine-tune MBart50 for Vietnamese-English medical translation.

This design evaluates LLM adaptability to domain-specific, low-resource tasks.

**Process
Design**

Materials and Tools

Datasets Utilized

- *French-English*: Global Voices and Europarl corpus (~3,000 sentence pairs)
- *Vietnamese-English*: MedEV dataset (~200,000 medical sentence pairs)

Libraries & Tools

- Hugging Face Transformers (MBart50, Seq2SeqTrainer)
- Tokenization: MBart50TokenizerFast, SentencePiece, MosesTokenizer
- Evaluation: Evaluate library for BLEU, METEOR, TER, and BERTScore
- Environment: Google Colab with A100 GPU support

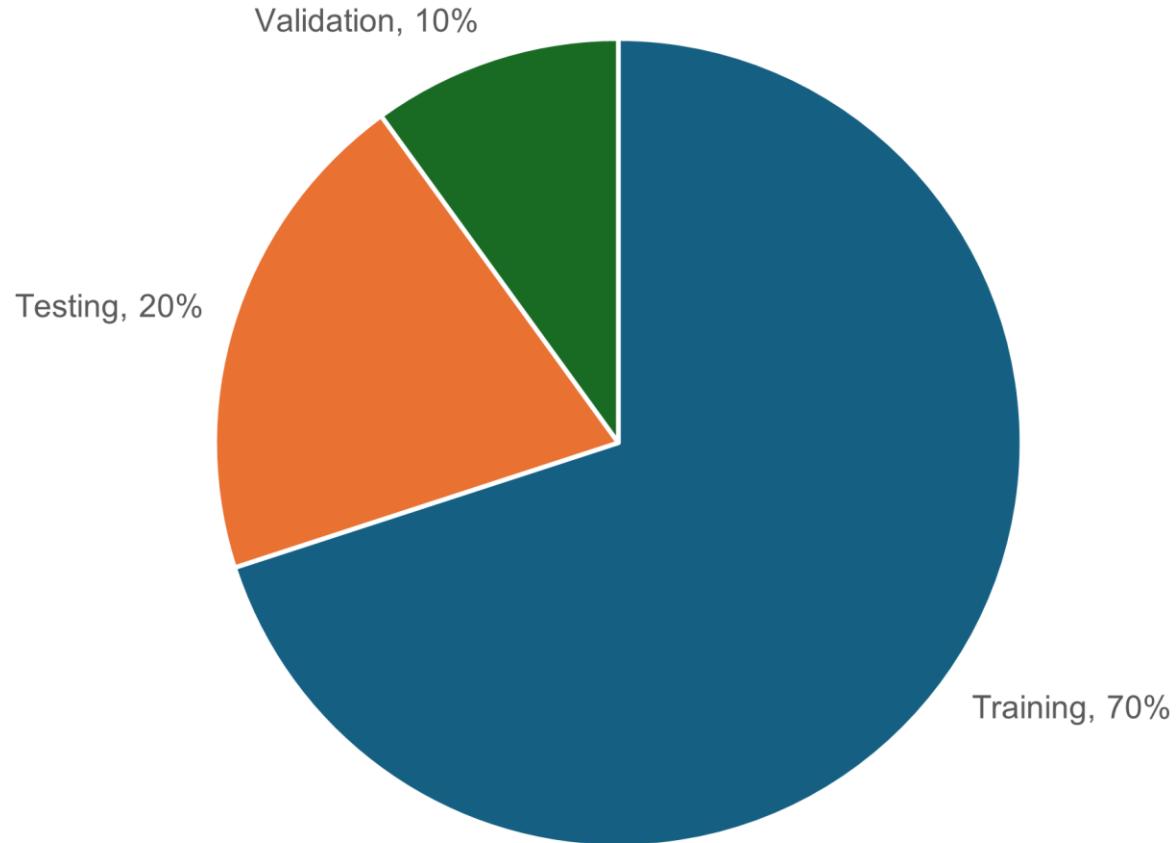
The main limitations include:

- Limited training from Google Colab's computational resources, which constrained batch size and training time
- Use of written parallel data does not fully reflect the nuances of spoken healthcare communication
- MBart50 offers broad multilingual support, but its vocabulary is not tailored to Vietnamese medical terminology. MBart50 and MBERT, while multilingual, are pretrained on uneven language distributions
- Low-resource languages like Vietnamese have limited annotated parallel corpora

Constraints and Limitations

03 Implementation

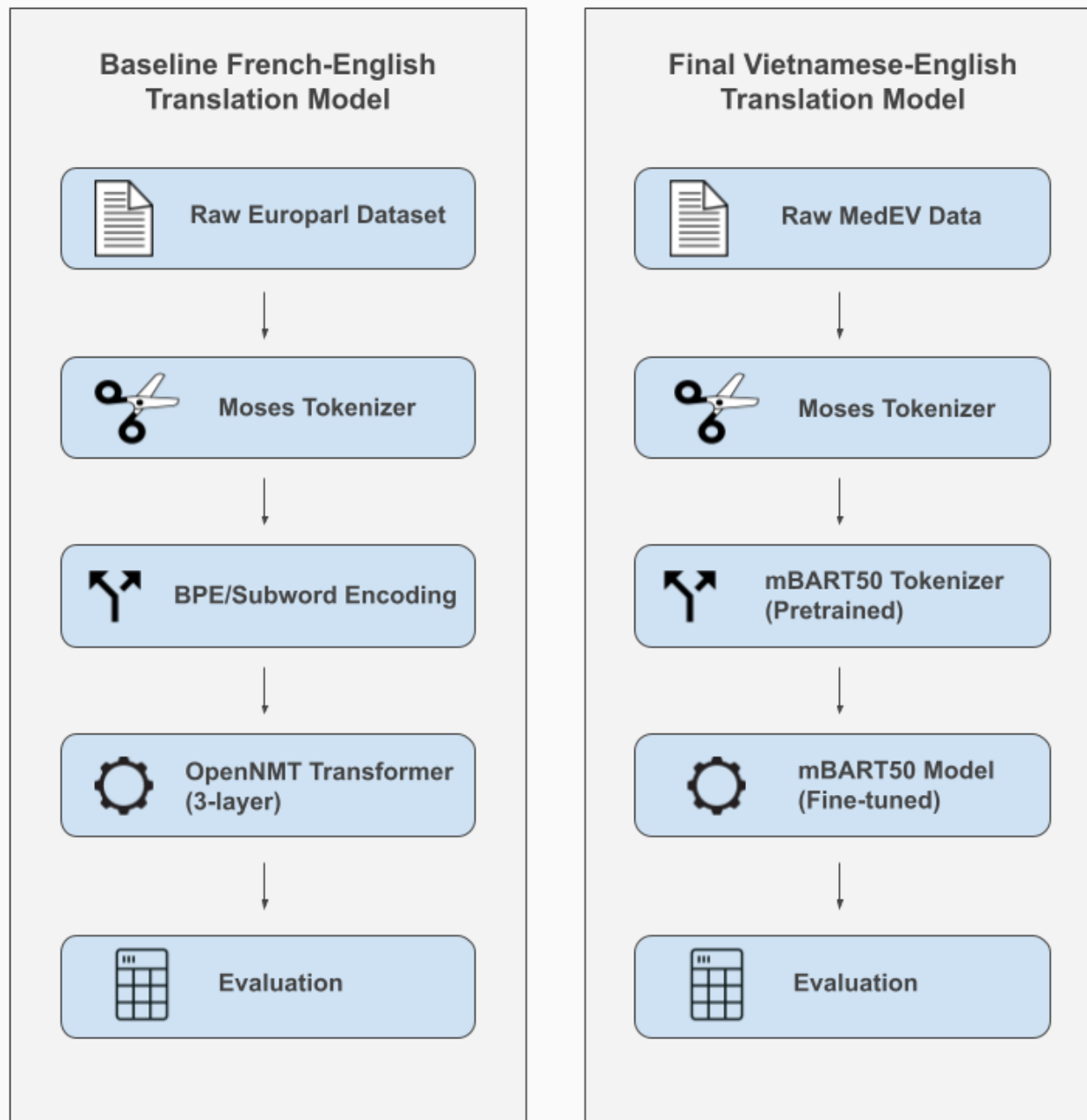
Dataset Splitting Ratio



Data

Preprocessing

- Cleaned and tokenized datasets using MosesTokenizer and SentencePiece/MBart50 tokenizer
- Split into training (70%), validation (10%), and testing (20%)

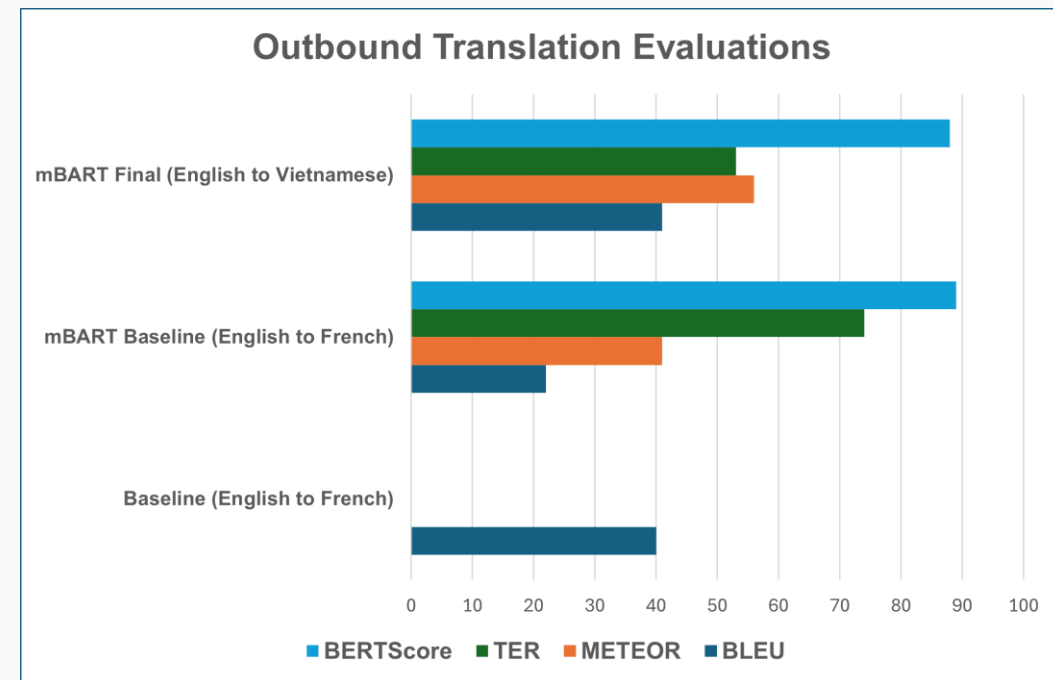
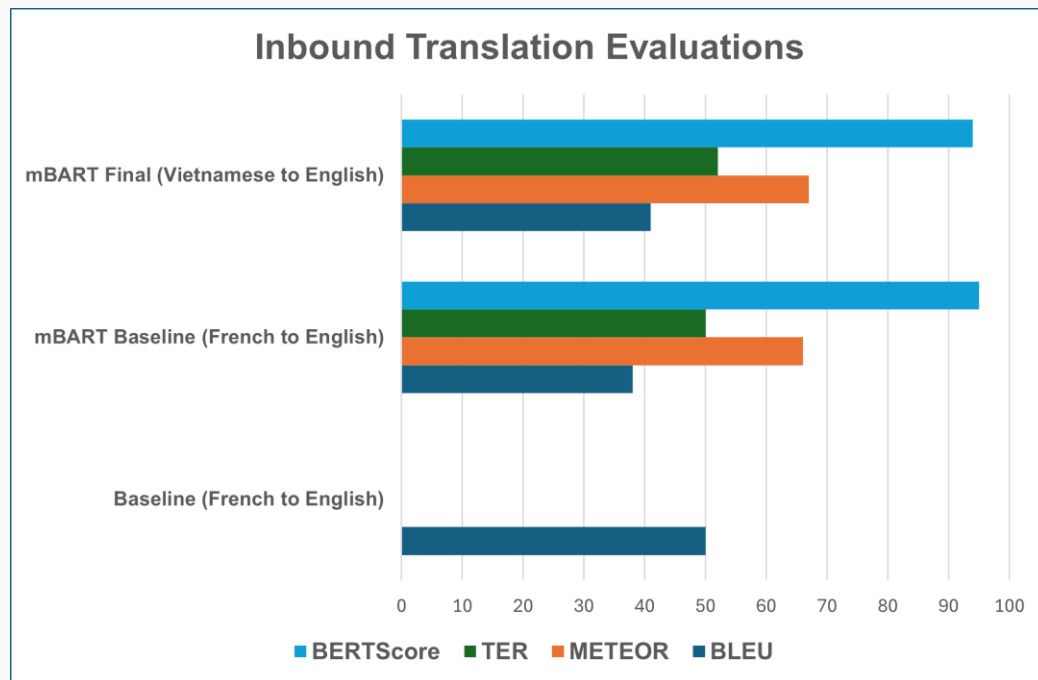


Model Development

1. **Baseline:** OpenNMT-py
Transformer models for French-English benchmarking
2. **Target Model:** MBart50 fine-tuned on MedEV using Seq2SeqTrainer with:
 1. 5-6 epochs, label smoothing (0.1), mixed-precision (fp16)
 2. Beam search (6 beams), early stopping

Evaluation: Assessed bidirectional performance using BLEU (precision), METEOR (semantics), TER (edit distance), and BERTScore (context)

04 Results



Inbound and Outbound Evaluation Scores

To evaluate translation quality, we used BLEU, METEOR, TER, and BERTScore. The final mBART Vietnamese-English model outperformed both the baseline and French-English models, benefiting from fine-tuning and a larger, domain-specific dataset.

Inbound Direction (Vietnamese → English):

BLEU: ~41, METEOR: ~67, TER: ~52, BERTScore: ~94

This indicates strong semantic and contextual alignment with human references.

Outbound Direction (English → Vietnamese):

BLEU: ~41, METEOR: ~56, TER: ~52, BERTScore: ~88

Final mBART Vietnamese-English Model Evaluation Metrics

Results Comparison

Compared to the French-English baseline, which achieved BLEU scores of ~50 (inbound) and ~40 (outbound), our Vietnamese model demonstrates that with targeted fine-tuning and LLM integration, low-resource languages can reach competitive translation quality.

Both evaluation graphs clearly illustrate that the integration of MBart50 with medical-domain fine-tuning significantly boosts translation performance, especially when compared to traditional NMT baselines.

05 Conclusion

Conclusion

This research demonstrates the potential of large multilingual pretrained models like MBart50 in addressing healthcare communication challenges for low-resource languages. By fine-tuning MBart50 on a domain-specific Vietnamese-English medical dataset, we achieved high translation performance across multiple evaluation metrics. Our final model outperformed traditional baselines and even French-English models in some metrics, confirming that with careful preprocessing, training, and evaluation, LLMs can effectively support underrepresented languages. These findings highlight the importance of integrating domain-specific data and targeted training strategies to build inclusive and reliable NMT systems for critical fields like the healthcare industry.

Future Work

To build on these results, future research should:

- Extend to other low-resource medical languages
- Integrate spoken dialogue and clinical data for better real-world performance
- Apply domain-adaptive pretraining or instruction tuning for medical relevance
- Develop lightweight, mobile-friendly models
- Conduct user-centered evaluations with healthcare professionals and native speakers

These directions will improve the practicality and accessibility of neural machine translation in multilingual healthcare settings.

References

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